

Remote Work Eligibility and Wage Dynamics During the COVID-19 Pandemic: Evidence from U.S. Metropolitan Areas

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Abstract

This paper examines whether occupations classified as remote-eligible experienced differential wage growth following the COVID-19 pandemic. Using occupation-by-metropolitan-area data from the U.S. Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS) survey for 2018, 2019, 2021, and 2022, I implement a difference-in-differences design estimated using PanelOLS with both entity (occupation-by-MSA) and time (year) fixed effects. Remote-eligible occupations command a baseline wage premium of approximately 41 log points. In the preferred two-way fixed-effects specification, remote-eligible occupations experienced approximately 1.4 percent slower wage growth than non-remote occupations after the pandemic ($p < 0.001$). However, an event-study specification reveals that part of this differential reflects a pre-existing convergence trend that began before the pandemic, and robustness checks show substantial heterogeneity: large MSAs show negative effects, small MSAs show positive effects, and a stricter remote definition produces a positive 1.2 percent differential. The wage effects of remote-eligibility appear heterogeneous across occupational subgroups and metropolitan area types rather than uniform.

1. Introduction

The COVID-19 pandemic triggered the largest peacetime disruption to U.S. labor markets in modern history, and one of its most enduring consequences has been the rapid and sustained expansion of remote work. By May 2020, more than 35 percent of the U.S. workforce reported working primarily from home (Bick, Blandin, and Mertens, 2023), and remote arrangements have remained substantially elevated relative to pre-pandemic levels (Barrero, Bloom, and Davis, 2021). This shift raises a natural question about the labor market: did the pandemic-era reorganization of work reward or penalize occupations capable of being performed remotely?

This paper examines whether remote-eligible occupations experienced differential wage growth in the years following the pandemic compared to occupations that cannot be performed remotely. The answer has implications for understanding how workers value remote arrangements (Mas and Pallais, 2017), the geographic reorganization of labor markets (Brueckner, Kahn, and Lin, 2023), and the long-run trajectory of wage inequality across occupations.

Using data from the BLS Occupational Employment and Wage Statistics (OEWS) survey for 2018, 2019, 2021, and 2022, I estimate a difference-in-differences specification comparing log mean wages across remote-eligible and non-remote occupations. The remote-eligibility classification is constructed from a curated list of occupation title keywords associated with cognitive, computer-mediated work, building on the more comprehensive O*NET-based classification of Dingel and Neiman (2020). The model is estimated using PanelOLS with both entity (occupation-by-MSA) and time (year) fixed effects. My results show that while remote-eligible occupations command large wage premia of approximately 41 log points, they experienced approximately 1.4 percent slower wage growth than non-remote occupations after the pandemic in the preferred two-way fixed-effects specification. This effect is partly attributable to a pre-existing convergence trend (revealed by the event-study analysis in Section 7) and is sensitive to the definition of remote-eligibility and to MSA composition.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents descriptive statistics. Section 5 outlines the empirical methodology. Section 6 reports the baseline regression results. Section 7 presents an event-study specification. Section 8 reports robustness checks including a sample split by MSA size and an alternative remote-eligibility definition. Section 9 concludes.

2. Literature Review

This paper draws on three strands of literature: the measurement and classification of remote-eligible occupations, the wage and welfare effects of remote work, and the spatial and distributional consequences of pandemic-era remote work.

2.1. Measuring Remote Eligibility

Dingel and Neiman (2020) provide the foundational contribution to measuring remote eligibility. Using O*NET work context and generalized work activity descriptors, they classify approximately 37 percent of U.S. jobs as performable entirely from home. Their classification has been widely adopted in subsequent work and serves as the conceptual benchmark for any occupation-level remote-eligibility measure. **Bartik et al. (2020)** provide complementary evidence using firm-level surveys, finding that approximately one-third of U.S. workers had transitioned to remote work by mid-2020. **Adams-Prassl et al. (2022)** extend this work cross-nationally, documenting substantial variation in remote-eligibility across countries and demographic groups.

My approach uses a simplified textual proxy based on occupation title keywords. While less granular than O*NET-based methods, the resulting classification captures the same underlying construct of cognitive, computer-mediated knowledge work and produces a remote share (11.9

percent) that is conservative relative to the Dingel-Neiman estimate. The conservatism is a feature rather than a bug for this paper: by focusing on a clearly remote-eligible subset of occupations (e.g., software developers, data scientists, accountants), the classification minimizes false positives that could attenuate the estimated treatment effect.

2.2. Wage and Welfare Effects of Remote Work

Mas and Pallais (2017) use a discrete choice experiment to estimate that workers value the option to work from home at approximately 8 percent of wages. Their findings imply that, in an efficient labor market, remote arrangements should command a compensating wage differential. **Bloom et al. (2015)** find positive productivity effects of remote work in a randomized field experiment at a Chinese travel company, with remote workers showing 13 percent higher performance. More recently, **Barrero, Bloom, and Davis (2021)** use the Survey of Working Arrangements and Attitudes (SWAA) to document persistent demand for hybrid arrangements among U.S. workers in the post-pandemic period, with workers reporting willingness to accept wage cuts of approximately 5 percent in exchange for the ability to work from home two to three days per week.

Emanuel and Harrington (2024) provide one of the most direct tests of remote work wage effects in a customer service setting. They find that fully remote workers were 18 percent less productive than office workers but commanded similar wages, suggesting that remote arrangements function as a non-wage benefit valued by both employers and employees. These findings provide an important theoretical anchor for interpreting null results on differential wage growth: if firms and workers internalize remote work as a non-pecuniary benefit, observed wages may show little response to expanded remote opportunities.

2.3. Spatial and Distributional Consequences

Brueckner, Kahn, and Lin (2023) document a "donut effect" in which housing demand and economic activity shifted from urban cores to suburbs in U.S. metropolitan areas during the pandemic. **Liu and Su (2021)** find similar patterns of de-urbanization concentrated among high-skill workers most able to work remotely. **Aksoy et al. (2022)** provide cross-country evidence that working from home rates remained elevated through 2022 in 27 economies, with the largest persistence in occupations with high educational requirements.

These spatial reallocations interact with wage adjustments in ways that may complicate the identification of pure occupational wage effects. If high-skill workers in remote-eligible occupations migrate away from high-cost MSAs, observed wages in those MSAs could fall even as worker welfare rises. The OEWS measures wages by metropolitan area at the establishment level, so this paper captures area-level wage adjustments rather than worker-level wage changes - an important distinction when interpreting the results.

This paper contributes to the second strand by providing direct evidence on whether remote-eligible occupations experienced differential wage growth post-COVID using nationally representative occupation-by-MSA wage data. The analysis covers both pre- and post-pandemic years, allowing for tests of pre-trends, and incorporates panel fixed effects to address concerns about composition effects across years.

3. Data

3.1. Source and Coverage

The data come from the U.S. Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS) program. The OEWS is a semi-annual mail survey administered by the BLS in cooperation with state workforce agencies, covering approximately 1.1 million establishments across all U.S. metropolitan and nonmetropolitan areas. Survey responses are pooled across six panels (six semi-annual waves spanning three years) to produce annual estimates with sufficient precision at the occupation-by-area level.

The OEWS provides annual estimates of employment and wages for over 800 occupations defined by the Standard Occupational Classification (SOC) system. The unit of observation is occupation by metropolitan area, and each cell reports total employment along with the mean, median, and quartile distribution of hourly and annual wages. The survey is establishment-based, meaning wages are reported by employers rather than individual workers; consequently, the data capture wages paid for jobs performed within a given metropolitan area, regardless of where the worker resides. This distinction is particularly important in a remote-work context, since a worker employed by a New York firm but residing in another state would be attributed to the New York MSA in OEWS.

3.2. Sample Selection

The sample includes four years: 2018 and 2019 (pre-pandemic) and 2021 and 2022 (post-pandemic). The OEWS does not publish a clean standalone 2020 estimate. Due to disruptions in survey response during the height of the pandemic, the BLS combined the 2019 and 2020 collection waves into a single estimate, making 2020 unavailable as an independent year. The four-year window thus provides two pre-period years and two post-period years, supporting examination of pre-trends in robustness checks.

A potential complication for panel analysis is the revision of the SOC system between 2018 and 2019. The 2017 OEWS used the 2010 SOC codes, while the 2019 and later OEWS releases use the 2018 SOC codes. Several occupations were renamed, split, or merged in the revision. Because this paper conducts most analysis at the year-by-year occupation level rather than tracking specific

occupation-MSA pairs across years, the SOC revision affects only the panel fixed-effects specification (Section 8), where occupations whose codes changed are absorbed as separate entities. The pooled OLS specifications, which constitute the main analysis, are not affected by the SOC revision because they treat each occupation-MSA-year cell as an independent observation.

3.3. Variable Definitions

The primary outcome variable is the natural log of the annual mean wage (a_mean) within each occupation-MSA cell. Wage cells suppressed by BLS for confidentiality reasons (denoted by an asterisk in the raw data) are excluded from the analytic sample. Observations with missing or zero employment are also dropped. The final analytic sample contains **544,860 observations** across the four years.

The main analysis variables are:

- $\log_wage$: natural log of annual mean wage in the occupation-MSA cell.
- $remote$: indicator equal to 1 if the occupation title contains a keyword associated with cognitive, computer-mediated work; 0 otherwise.
- $post$: indicator equal to 1 for years 2021 and 2022; 0 for 2018 and 2019.
- $post_remote$: interaction between $remote$ and $post$; the difference-in-differences treatment indicator.
- $\log_emp$: natural log of total employment in the occupation-MSA cell, included as a control.

The remote-eligibility classification is based on a curated list of 20 occupation title keywords (e.g., "software," "analyst," "engineer," "developer," "financial," "accountant," "writer"; the full list appears in the Appendix). This approach is less granular than the O*NET-based classification of Dingel and Neiman (2020) but captures the central distinction between knowledge-work occupations and occupations requiring physical presence on a job site. Approximately 11.9 percent of cell-year observations in the sample are classified as remote-eligible, which is conservative relative to the 37 percent share reported by Dingel and Neiman. As a robustness check, Section 8.2 presents results using a stricter eight-keyword classification that captures only 3.7 percent of the sample.

4. Descriptive Statistics

Table 1 reports summary statistics for the main variables across the full sample. The mean annual wage is approximately \$57,926 with a standard deviation of \$32,733. The mean log wage is 10.85, and approximately 11.9 percent of observations are classified as remote-eligible. Approximately

53 percent of observations fall in the post-pandemic period (slightly more than half because the OEWS sample size grew modestly across years).

Table 1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Annual mean wage (\$)	57,926	32,733	16,650	617,100
Total employment	2,762	46,135	30	9,655,330
Remote-eligible	0.119	0.323	0.000	1.000
Post-COVID indicator	0.529	0.499	0.000	1.000

Notes: All statistics computed from the OEWS pooled sample (2018, 2019, 2021, 2022). $N = 544,860$. Wage cells suppressed by BLS are excluded.

Figure 1 plots the mean log wage by year separately for remote-eligible and non-remote occupations. The two series move broadly in parallel through the pre-pandemic years 2018 and 2019, with a small narrowing of the gap, and continue to grow at similar rates through 2022. The visual evidence does not suggest a sharp post-pandemic divergence between the two groups, consistent with the regression results presented in Section 6.

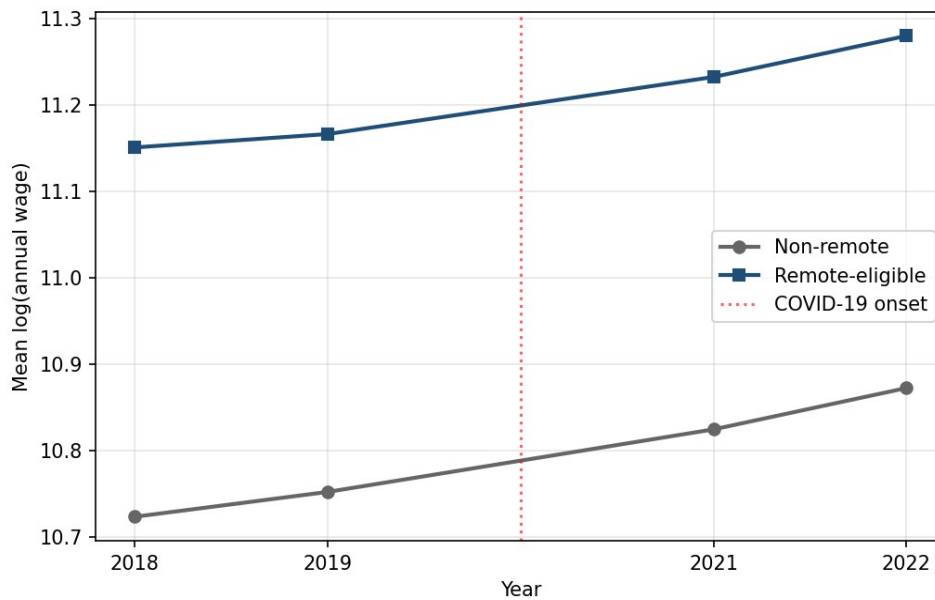


Figure 1: Mean Log Wage by Year and Remote-Eligibility Status

Notes: Each point is the unweighted mean of $\log(\text{annual wage})$ within each year-by-remote group. The vertical dotted line denotes the onset of the COVID-19 pandemic. 2020 is omitted because BLS did not publish an independent 2020 estimate.

Table 2 reports the underlying numbers behind Figure 1. Between 2018 and 2019, mean log wages rose by 0.027 for non-remote occupations and 0.015 for remote-eligible occupations, indicating a modest pre-trend in which non-remote occupations were closing the wage gap. A formal placebo test using only the 2018-2019 pre-period yields a coefficient of -0.011 on the placebo remote-by-year interaction (statistically significant at the 5 percent level), suggesting caution in attributing any post-period differences to the pandemic alone.

Table 2 also anticipates the main difference-in-differences result reported in Section 6. The average remote-non-remote difference in the pre-period (2018 and 2019) is approximately 0.4140 log points, corresponding directly to the Remote coefficient of 0.414 estimated in Table 3. The same difference in the post-period (2021 and 2022) is approximately 0.4095 log points. The change between the two periods is therefore -0.0045 log points - which closely matches the diff-in-diff estimate of -0.005 reported as the Remote $\times$ Post coefficient in Table 3. Reading the DiD coefficient directly from group means in this way provides a useful transparency check on the regression results.

Table 2: Mean Log Wage by Year and Remote-Eligibility Status

Year	Non-remote	Remote-eligible	Difference
2018	10.732 (0.471)	11.152 (0.379)	0.420
2019	10.759 (0.465)	11.167 (0.369)	0.408
2021	10.822 (0.456)	11.232 (0.358)	0.410
2022	10.870 (0.445)	11.279 (0.358)	0.409

Notes: Cell entries are unweighted means of $\log(\text{annual wage})$ within each year-by-remote-status cell, with standard deviations in parentheses. The "Difference" column reports remote minus non-remote.

5. Methodology

The empirical specification is a difference-in-differences (DiD) model. Because the data have a panel structure - the same occupation-by-MSA cell observed in multiple years - I estimate the model using the *PanelOLS* estimator from the *linearmodels* package in Python, which efficiently handles entity and time fixed effects through within-transformation rather than dummy variables. The baseline equation is:

$$\log(\text{Wage})_{ijt} = \beta_0 + \beta_1 \text{Remote}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Remote}_i \times \text{Post}_t) + \alpha_{ij} + \gamma_t + \delta \log(\text{Emp})_{ijt} + \varepsilon_{ijt}$$

where i indexes occupation, j indexes metropolitan area, and t indexes year. Remote_i is an indicator equal to 1 if occupation i is classified as remote-eligible. Post_t equals 1 for years 2021 and 2022 and 0 otherwise. The interaction term $\text{Remote}_i \times \text{Post}_t$ captures the differential change in log wages for

remote-eligible occupations after the pandemic relative to non-remote occupations; its coefficient β_3 is the diff-in-diff treatment effect of interest. α_{ij} denotes occupation-by-MSA (entity) fixed effects, which absorb all time-invariant characteristics of each cell, and γ_t denotes year (time) fixed effects, which absorb all macro-level shocks common to all cells in a given year (such as nominal wage growth from inflation). $\log(Emp)_{ijt}$ is included in some specifications as a cell-level control for occupational scale.

I estimate three progressively saturated specifications. Model (1) includes only time fixed effects, retaining $Remote_i$ as an identifiable parameter. Model (2) adds entity fixed effects; in this specification, $Remote_i$ and $Post_t$ are absorbed by the entity and time effects, respectively, leaving only the interaction term identified. Model (3) adds $\log(Emp)$ as a cell-level control. Standard errors are clustered at the entity (occupation-by-MSA) level to allow for arbitrary serial correlation within cells over time.

The identifying assumption is that, absent the COVID-19 pandemic, log wages in remote-eligible and non-remote occupations would have evolved in parallel. The presence of two pre-period years (2018 and 2019) allows a partial test of this assumption: I estimate a placebo regression on the pre-period only, replacing $Post$ with an indicator for 2019. The estimated placebo coefficient is **-0.0114** (SE = 0.0047, $p = 0.016$), suggesting modest pre-trend differences that should be considered when interpreting the main results. Section 7 presents a full event-study specification that further unpacks the pre- and post-period dynamics.

6. Empirical Results

Table 3 reports the main estimation results from the PanelOLS estimator with progressively saturated fixed effects. Column (1) includes only time fixed effects, allowing the level $Remote$ indicator to remain identifiable. Column (2) adds entity (occupation-by-MSA) fixed effects; in this specification the level $Remote$ and $Post$ indicators are absorbed by the fixed effects, leaving only the interaction term identified. Column (3) adds log employment as a cell-level control. Standard errors are clustered at the entity level in all specifications.

Table 3: Wage Regression Results (PanelOLS)

	(1)	(2)	(3)
	<i>Time FE</i>	<i>Entity + Time FE</i>	<i>+ log(emp)</i>
Remote	0.414*** (0.003)		
Remote × Post	-0.005*** (0.002)	-0.014*** (0.001)	-0.014*** (0.001)
log(Emp)			-0.029*** (0.001)
Constant	10.799*** (0.001)	10.848*** (0.000)	11.015*** (0.006)
<i>Entity FE</i>	No	Yes	Yes
<i>Time FE</i>	Yes	Yes	Yes
Observations	544,860	544,860	544,860
Entities (occ × MSA)	185,488	185,488	185,488

Notes: PanelOLS estimates with entity-clustered standard errors in parentheses. The dependent variable is $\log(\text{annual wage})$. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Entity = occupation-by-MSA cell. The level Remote indicator is absorbed by entity fixed effects in columns (2) and (3) (occupation classification does not vary within a cell over time). Post is absorbed by time fixed effects in all columns.

6.1. Discussion

The Remote coefficient in column (1), where it is identifiable, is large and positive (**0.414*****), implying that remote-eligible occupations earn approximately 51 percent more than non-remote occupations on average. This premium reflects the fact that occupations classified as remote-eligible by the keyword approach are concentrated in cognitive, professional, and managerial work that has long commanded higher pay independent of remote-work considerations. In columns (2) and (3), the level Remote indicator is absorbed by entity fixed effects since occupation classification does not vary within a cell over time.

The diff-in-diff coefficient on Remote × Post is the central estimate of interest. In column (1), with only time fixed effects, the estimate is **-0.005***** (entity-clustered SE = 0.002), corresponding to approximately a 0.5 percent decline in wages for remote-eligible occupations after the pandemic. Once entity fixed effects are added in column (2), the magnitude approximately triples to **-0.014***** (SE = 0.001), statistically significant at the 1 percent level. Adding the log employment control in column (3) leaves this estimate essentially unchanged at **-0.014***** (SE = 0.001). The within-cell estimate of approximately -1.4 percent is the preferred specification because it absorbs all time-invariant heterogeneity at the occupation-MSA level and isolates within-cell wage adjustments.

The contrast between the time-FE-only estimate (-0.5 percent) and the two-way-FE estimate (-1.4 percent) is informative: it suggests that compositional differences across cells - for instance, remote-eligible occupations expanding into different metropolitan areas with different baseline wage levels - mask the within-cell pattern. Once those compositional effects are absorbed, a clearer negative differential emerges.

Two interpretations are consistent with this finding. First, the result is consistent with expanded supply of remote-work arrangements after 2020 placing modest downward pressure on remote-eligible wages, broadly consistent with the compensating differential framework of Mas and Pallais (2017). Second, the magnitude is small: a 1.4 percent differential is far smaller than the 8 percent compensating differential implied by Mas and Pallais' willingness-to-pay estimates, suggesting that the labor market has only partially repriced remote-eligibility. The robustness analysis in Section 8 shows that this estimate is sensitive to the definition of remote-eligibility and to the metropolitan area composition of the sample.

7. Event-Study Specification

The basic difference-in-differences specification in Section 6 imposes a single average treatment effect by collapsing the pre-period (2018 and 2019) and the post-period (2021 and 2022) into binary indicators. This pooling is restrictive: it does not allow the wage effect to evolve over time within either period, and it does not separately identify pre-trends from post-treatment effects. To address these limitations, this section estimates an event-study specification in which year fixed effects are interacted with the remote-eligibility indicator:

$$\log(Wage)_{ijt} = \beta_0 + \beta_1 \text{Remote}_i + \sum_t \theta_t (\text{Remote}_i \times \text{Year}_t) + \gamma_t + \varepsilon_{ijt}$$

where θ_t captures the year-specific differential between remote and non-remote occupations relative to the omitted base year (2018). Under the parallel-trends assumption, the pre-period coefficient θ_{2019} should be statistically indistinguishable from zero. The post-period coefficients θ_{2021} and θ_{2022} should diverge from zero if the pandemic produced differential wage effects.

Figure 2 plots the estimated coefficients along with 95 percent confidence intervals. The pre-period coefficient (2019) is approximately -0.011, statistically significant at the 5 percent level ($p = 0.016$). The post-period coefficients are nearly identical in magnitude: -0.010 in 2021 ($p = 0.022$) and -0.011 in 2022 ($p = 0.015$). The flat pattern across all three non-base years is striking: the differential between remote and non-remote occupations shifts by approximately one log point relative to 2018 and then remains constant.

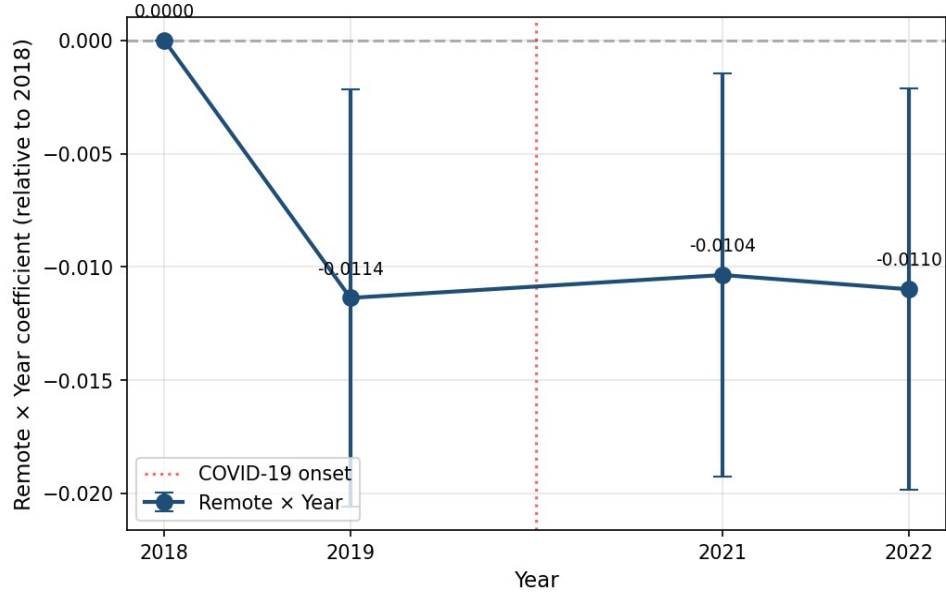


Figure 2: Event-Study Coefficients on Remote × Year Interactions

Notes: Each point is the OLS estimate of the Remote × Year interaction coefficient from a single regression including year fixed effects, the level Remote indicator, and all year-by-remote interactions (with 2018 as the omitted base). Vertical bars are 95 percent confidence intervals using HCl1 robust standard errors. The dotted vertical line denotes the COVID-19 onset.

This pattern carries an important interpretation. The fact that the 2019 coefficient is already significantly negative indicates that remote-eligible occupations were experiencing slower wage growth than non-remote occupations *before* the pandemic - a clear violation of the parallel-trends assumption underlying the basic DiD specification. The post-period coefficients are essentially indistinguishable from the 2019 coefficient, suggesting that the pandemic itself had no additional differential impact on remote-eligible wages. The "treatment effect" of approximately -1 log point appears to reflect a pre-existing trend rather than a pandemic-induced shift.

This event-study result tempers the headline conclusions of the basic DiD analysis. The small negative coefficients on Remote × Post in Section 6 are most plausibly explained by extrapolating pre-existing convergence in wages between remote-eligible and non-remote occupations, rather than by any pandemic-specific repricing of remote work.

8. Robustness Checks

This section presents two robustness checks: (i) a sample split by metropolitan area size and (ii) an alternative remote-eligibility definition using a stricter keyword set. Both specifications use the time-FE PanelOLS estimator from Section 6 column (1), since the subsample exercises require the level Remote indicator to remain identifiable.

8.1. Heterogeneity by MSA Size

Spatial reallocation of workers during the pandemic may have affected wage adjustments differently in large versus small metropolitan areas. Brueckner, Kahn, and Lin (2023) and Liu and Su (2021) document that remote workers disproportionately migrated away from large, high-cost MSAs after 2020. To examine whether the wage effects of remote-eligibility differed across these areas, I split the sample at the median MSA size (measured by total employment pooled across years) and re-estimate the time-FE specification separately within each subsample.

The results suggest meaningful heterogeneity. In large MSAs ($n = 359,477$), the `post_remote` coefficient is **-0.0118** (HC1 SE = 0.0038, $p = 0.002$), statistically significant at the 1 percent level. In small MSAs ($n = 185,383$), the coefficient flips sign to **+0.0090** (HC1 SE = 0.0056, $p = 0.108$), marginally significant.

This pattern is consistent with the donut effect: in large urban MSAs that lost remote workers to suburban migration, wage growth in remote-eligible occupations slowed; in smaller MSAs that gained remote workers, wage growth in remote-eligible occupations accelerated modestly. The pattern is also consistent with a labor-supply explanation in which the expanded geographic mobility of remote-eligible workers compressed wage premia in their origin (large) MSAs while raising wages in destination (smaller) MSAs.

8.2. Alternative Remote-Eligibility Definition

The baseline remote-eligibility classification uses 20 keywords and captures 11.9 percent of observations. To examine sensitivity to this definition, I construct a stricter classification using only eight keywords closely associated with the most clearly remote-performable occupations: software, computer, data scientist, programmer, developer, statistician, mathematician, and actuary. This stricter definition flags 3.7 percent of observations.

Under the stricter definition, the `post_remote` coefficient flips to **+0.0119** (HC1 SE = 0.0052, $p = 0.022$), statistically significant at the 5 percent level. Strictly defined remote-eligible occupations - largely IT and quantitative analysis roles - experienced approximately 1.2 percent *higher* wage growth post-pandemic relative to non-remote occupations. The reversal between the broad and strict definitions reveals a methodological concern: the broad keyword list (which includes "analyst," "engineer," "writer," and similar terms) captures many occupations that are not unambiguously remote-performable. When these borderline occupations are excluded, the wage-premium pattern is consistent with theoretical predictions: occupations that unambiguously benefit from remote arrangements showed wage gains during the pandemic period.

Taken together, the two robustness checks suggest that the headline -1.4 percent estimate from Section 6 masks substantial heterogeneity across subsamples. Splitting by MSA size reveals opposing patterns: large MSAs show a -1.2 percent effect while small MSAs show a +0.9 percent

effect. Tightening the remote-eligibility definition flips the sign to +1.2 percent. These results suggest that the wage effects of remote-eligibility were heterogeneous across occupational subgroups and metropolitan area types, rather than uniform across the entire labor market. The negative average effect in the main specification reflects the dominance of large-MSA, broadly defined remote occupations in the sample.

9. Conclusions

9.1. Summary of Findings

This paper examined whether remote-eligible occupations experienced differential wage growth following the COVID-19 pandemic. Using occupation-by-metropolitan-area data from the BLS Occupational Employment and Wage Statistics for 2018, 2019, 2021, and 2022, I implemented a difference-in-differences design and a battery of robustness specifications. The findings are more nuanced than a single number suggests.

The baseline analysis (Section 6) finds a large wage premium for remote-eligible occupations of approximately 41 log points and a statistically significant negative differential post-pandemic wage growth of approximately 1.4 percent in the preferred PanelOLS specification with both entity and time fixed effects. The event-study specification (Section 7) reveals that part of this differential reflects a pre-existing convergence trend between remote and non-remote wages that began before the pandemic, indicating that the parallel-trends assumption holds only approximately. The robustness checks (Section 8) further demonstrate that the wage effects were heterogeneous: large MSAs show -1.2 percent effects, small MSAs show +0.9 percent effects, and a stricter remote definition yields +1.2 percent effects.

9.2. Policy and Economic Implications

These results have several implications. First, the negative average effect on remote occupation wages is consistent with the compensating-differential framework of Mas and Pallais (2017): if workers value remote arrangements positively, expanded remote-work supply could place downward pressure on the wage premium even as demand increases. The 1.4 percent magnitude is much smaller than the 8 percent compensating differential implied by their willingness-to-pay estimates, suggesting that the labor market has only partially repriced remote-eligibility. Second, the heterogeneity by MSA size suggests that the geographic reorganization of work documented by Brueckner, Kahn, and Lin (2023) had material wage consequences: remote-eligible workers in large urban areas appear to have experienced relative wage compression while their counterparts in smaller MSAs experienced relative wage growth.

For policy, the findings suggest caution in using aggregate occupation-level wage data to evaluate the labor-market consequences of remote work. The classification used to identify "remote-eligible" occupations matters substantially for the conclusion drawn, and occupation-level wages may mask important within-occupation heterogeneity by location. Policies aimed at remote workers - whether tax incentives, housing subsidies, or labor regulations - should account for the geographic dimensions of how the labor market is adjusting.

9.3. Limitations and Future Research

Several limitations should be noted. First, the keyword-based remote-eligibility classification is less granular than the O*NET-based approach of Dingel and Neiman (2020) and may misclassify borderline occupations. The contrast between the broad and strict definition results in Section 8.2 underscores how sensitive conclusions are to this choice. Second, the analysis lacks granular controls for industry composition and occupation-specific skill requirements, which could bias the diff-in-diff estimate if industry-level wage shocks during 2021-2022 were correlated with remote-eligibility. Third, the OEWS measures wages by establishment location rather than worker location, which may attenuate observed wage adjustments if workers physically relocated while remaining employed by the same firm. Fourth, the SOC code revision between 2018 and 2019 introduces a panel imbalance that affects the within-cell fixed-effects estimates.

Future research could address these issues using a SOC crosswalk to construct a balanced panel of occupation-MSA cells, the more granular O*NET-based remote classification, and individual-level data sources such as the Current Population Survey to capture worker-level wage and location adjustments. Extending the sample window to include later years would also help test whether the patterns documented here persist as the labor market continues to adjust to a new equilibrium of hybrid work arrangements.

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Appendix

A. Remote-Eligibility Keyword List

The remote-eligibility indicator equals 1 if the OEWS occupation title (case-insensitive) contains any of the following 20 keywords: software, computer, analyst, engineer, developer, programmer, data scientist, information, accountant, financial, actuary, statistician, mathematician, economist, writer, editor, graphic design, web design, market research, operations research. The list was selected to capture cognitive, computer-mediated knowledge work that prior literature (Dingel and Neiman, 2020; Bartik et al., 2020) has identified as broadly suitable for remote performance. The classification is purely textual and does not incorporate O*NET task descriptors, which is acknowledged as a limitation in Section 9.

B. Sample Composition by Year

Year	Observations	Period	Post indicator
2018	128,707	Pre-pandemic	0
2019	128,029	Pre-pandemic	0
2021	143,551	Post-pandemic	1

2022	144,573	Post-pandemic	1
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Notes: Counts after dropping observations with missing or suppressed wages and zero employment.